Evidence Preservation and Analysis Report

1. Introduction

Evidence preservation is a critical phase of incident response in a Security Operations Center (SOC). It ensures that digital evidence is collected, stored, and analyzed in a forensically sound manner while maintaining data integrity and legal admissibility. Proper evidence handling allows security teams to reconstruct attack scenarios, identify threat actors, and support legal or compliance investigations.

In this task, volatile and non-volatile evidence was collected from a simulated compromised Windows virtual machine using forensic tools such as **Velocidex Enterprises Velociraptor** and **AccessData FTK Imager**. Network connection data and memory artifacts were acquired, preserved, and documented using a structured chain-of-custody process.

This exercise demonstrates best practices in digital evidence acquisition, preservation, and analysis within a controlled SOC environment.

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2. Objective

The objective of this task is to collect, preserve, and analyze digital evidence from a compromised system while maintaining data integrity and proper documentation.

Key Objectives

- Collect volatile system data.
- Acquire memory artifacts from a compromised host.
- Maintain evidence integrity using hashing.
- Document chain-of-custody procedures.
- Preserve evidence for forensic investigation.

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3. Tools Used

- Velociraptor - Live forensic data collection
- FTK Imager - Memory acquisition and evidence imaging

4. Evidence Preservation Overview

Digital evidence preservation involves collecting system data without modifying its original state. Evidence must remain reliable and verifiable throughout the investigation process.

Types of Digital Evidence

- **Volatile Evidence** — Memory and active network connections.
- **Non-Volatile Evidence** — Disk data and system logs.
- **Forensic Artifacts** — System processes and runtime information.

Proper handling ensures that evidence remains admissible and trustworthy.

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5. Methodology / Implementation

5.1 Volatile Data Collection

Volatile data represents information that exists temporarily in system memory and is lost when the system shuts down. Therefore, it must be collected immediately during incident response.

Network connection data was collected from the Windows virtual machine using Velociraptor.

Executed Query

```
SELECT * FROM netstat
```

This query retrieved:

- Active network connections
- Local and remote IP addresses
- Port numbers
- Connection states

The collected data was exported and saved in CSV format for analysis.

5.2 Memory Acquisition

Memory acquisition was performed to capture runtime system information that may contain evidence of malicious activity.

Memory Collection Process

- Acquired memory image from compromised system.
- Created forensic copy of memory data.
- Stored evidence in a secure location.

Memory artifacts may contain:

- Running processes
- Malware traces
- User credentials
- System execution history

5.3 Evidence Integrity Verification

To ensure that collected evidence remained unchanged, a cryptographic hash value was generated.

Hashing Procedure

- Generated SHA-256 hash of memory dump.
- Recorded hash value for verification.
- Used hash comparison to confirm data integrity.

Hashing ensures that evidence has not been modified during storage or analysis.

5.4 Evidence Documentation

All collected evidence was documented to maintain accountability and traceability.

Evidence Collection Record

Item	Description	Collected By	Date	Hash Value
Memory Dump	Windows VM Memory Image	SOC Analyst	2025-08-18	SHA256-A1B2C3D4E5F67890

5.5 Chain-of-Custody Documentation

Chain-of-custody records track how evidence is handled throughout the investigation lifecycle.

Chain-of-Custody Log

Date	Time	Action Performed	Evidence Item	Responsible Person
19-02-2026	16:10	Identified system as compromised	Windows VM	SOC Analyst
19-02-2026	16:15	Collected network connection data	Netstat Output	SOC Analyst
19-02-2026	16:20	Acquired memory dump	Memory Image	SOC Analyst
19-02-2026	16:25	Generated SHA-256 hash	Memory Image	SOC Analyst
19-02-2026	16:30	Stored evidence securely	Forensic Storage	SOC Analyst

Maintaining chain-of-custody ensures evidence authenticity and accountability.

6. Evidence Analysis

Analysis of collected evidence provided insights into system activity.

Key Observations

- Active external network connections were identified.
- Suspicious remote connections indicated possible compromise.
- Memory artifacts preserved runtime system behavior.
- Evidence supported further forensic investigation.

The collected data helps identify attack origin and system impact.

7. Security Impact

Proper evidence preservation supports:

- Accurate incident investigation.
- Legal and compliance requirements.
- Identification of attack techniques.
- Root cause analysis.
- Improved security response strategies.

Failure to preserve evidence correctly may compromise investigation outcomes.

8. Conclusion

This task demonstrated the process of digital evidence preservation and forensic data collection within a SOC environment. Volatile data and memory artifacts were collected, secured, and verified using cryptographic hashing while maintaining proper chain-of-custody documentation. Effective evidence preservation ensures reliable forensic analysis and supports incident response, threat investigation, and organizational security improvement.